MIL-P-46987B(MU)
13 DECEMBER 1965
SUPERSEDING
MIL-P-46987A(MU)
23 NOVEMBER 1964

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, CHEMICAL AGENT VX, M426, WITHOUT FUZE AND BURSTER; FILLING OF

1. SCOPE

1.1 This specification covers the filling of an 8 inch projectile with VX.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-V-13750 — Varnish, Phenol-Formaldehyde, Clear and Aluminum Pigmented.

MIL-D-51083 — Detector Ticket, Anticholinesterase Agent.

MIL-S-51189 — Substrate Solution and Dispenser.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

PICATINNY ARSENAL

F7548346 — Packing and Marking of Pallet for 8-Inch Projectile, Empty or Loaded.

F8860622 — Projectile, 8-Inch, Gas, Persistant, VX, M426, Filling Assembly.

(All drawings referenced in this specification are listed on Drawing F8860622.)

FSC 1320

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PUBLICATIONS

AMMUNITION PROCUREMENT AND SUPPLY AGENCY

QAI 14-210-1 — Preparation of Ammunition Data Cards.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain, prior to their use, that all materials and components have met their respective specifications. The agent shall meet the requirements of the specification under which the agent was manufactured.

3.2 Rust prevention of components. The rust prevention protection on the projectile and components shall not be removed until just prior to filling.

3.3 Filling, closing and marking. The projectile, clean and free from corrosion and foreign matter, shall be filled with agent (see 6.2) and closed and marked in accordance with Drawing F8860622. The projectile shall be kept in a nose-up position after filling until completion of test specified in 4.6.2 (see 6.3).

3.4 Agent contamination.

3.4.1 *Projectile.* The projectile shall be free from contamination when tested as specified in 4.6.1.1.

3.4.2 *Closure.* The closure shall be free from contamination when tested as specified in 4.6.1.2.

3.5 Helium leakage. The projectile closure shall not leak helium at a rate greater than 1 x 10^{-5} atmospheric cubic centimeter per second when tested as specified in 4.6.2.

3.6 Filling level. The liquid level in the filled and closed projectile shall be within the limits shown in table I when tested as specified in 4.6.3.

TABLE I. *Filling level*

Temperature range °F	*h	
	Minimum	Maximum
Between 40 and 50	7.21	7.96
50 and 60	7.12	7.87
60 and 70	6.99	7.79
70 and 80	6.86	7.68
80 and 90	6.73	7.55
90 and 100	6.62	7.45
100 and 110	6.53	7.35

* Inches below nose of projectile.

Calibration factors:

9 percent void at 40°F in a minimum munition.

Expansion factor VX — 0.00046 ml/°F — ml agent.

3.7 Burster case coating. Any protective coating on the interior of the burster case removed during decontaminating process shall be corrected. The varnish shall conform to grade A, type I of MIL–V–13750, and shall be a color similar to that on the interior of the burster casing. After the burster casing has been revarnished the bore shall accept freely a plug gage of 2.650 — 0.000 + 0.002 inch outside diameter by 25.86 inch minimum length.

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of projectiles shall be produced in accordance with this specification for examination and tests (see 4.2.3).

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3.9 Workmanship. The projectiles **shall** be free from visible damage such as dents, cracks, or deformation that could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.1.1 *Filling level.* When an alternative procedure is approved for use in determining the filling level, the procedure **shall** be monitored in accordance with 4.6.3 to insure that the requirements of this specification are satisfied.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Preproduction.* A sample of fifty filled projectiles shall be produced using the same materials, methods, equipment, process and procedure as will be used in production. Any change in materials, methods, process or procedure will require a new preproduction sample. The acceptance number is zero.

4.3 Lotting. A lot shall consist of the projectiles filled and assembled by one manufacturer, at one plant, using the same materials, under essentially the same filling conditions. However, no more than one lot of agent, one lot of metal parts as identified by one manufacturer and one metal part interfix number shall be represented in any one lot of filled projectiles. For ballistics considerations a sublot shall consist of the projectiles having the same zone designation (see 5.1).

4.4 Sampling.

4.4.1 *For examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *For filling level.* Sampling shall be conducted in accordance with MIL–STD–1235.

4.5 Inspection provisions.

4.5.1 *For examination.* Sample projectiles and the level A preparation for delivery shall be examined in accordance with the classification of defects and MIL–STD–105.

4.5.2 *Filling level.* The inspection for filling level shall be in accordance with 4.6.3 and CSP–1 of MIL–STD–1235 using an AQL of 0.25 percent defective.

4.5.3 *For decontaminating process.* Fifty projectiles from this first shift produced under regular production conditions shall be tested as specified in 4.6.1 and 4.6.1.1 to qualify the decontaminating process or whenever the decontaminating process is modified.

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4.5.4 *Closure contamination.* At the beginning of each work shift, five consecutive projectiles shall be tested for contamination in accordance with 4.6.1.2. If none is contaminated, every fifty projectile thereafter shall be sampled. If one or more of the first five is found to be contaminated, the next 10 consecutive projectiles produced in the shift shall be tested. If all are found uncontaminated. every fifty projectile thereafter shall be tested. If one or more of the 10 is found to be contaminated, the entire shift production up to that time shall be recycled through the decontaminating process and retested for contamination as above. Whenever a projectile is found to be contaminated during sample inspection, the 10 projectiles preceding and following the contaminated projectiles shall be tested for contamination. If all the projectiles are found to be uncontaminated, every fifth projectile shall be tested. If one or more of the projectiles are found to be contaminated, the entire quantity produced in the shift shall be recycled through the decontaminating process.

4.5.5 *For critical defects.* Each projectile in the lot shall be inspected for critical defects listed in the classification of defects.

4.5.6 *Classification of defects.*

4.5.6.1 *Projectile, 8-inch, M426, VX, filling assembly (Dwg F8860622).*

Categories	*Defects*	*Acceptance standards*
Critical:		
1	Helium leakage excessive	4.6.2
2	Zone weight marking incorrect by two or more zones	4.6.4
3	Marking misleading	
4	Color code incorrect	
Major:	AQL 1.0 percent defective	
101	Component missing	
102	Component incorrectly assembled	
103	Bourrelet diameter over max after painting	
104	Threads damaged (3.9)	
105	Rotating band damaged (3.9)	
106	Depth from face to top of burster casing incorrect	
107	Coating missing from adapter threads	
108	Torque test failure of adapter	
109	Varnish removed from interior of burster casing	
110	Lubricant missing from press fit surface (white lead)	
111	Any weight zone punch mark inadequate	
Minor:	AQL 2.5 percent defective	
201	Marking missing	
202	Exterior paint missing	

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4.5.6.2 *Preparation for delivery (section 5).*

Categories	*Defects*	*Acceptance standards*
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Strapping loose	
102	Strapping missing	
103	Pallet broken	
104	Marking missing	
105	Grommet missing	
106	Shipping gasket missing	
107	Mixed weight zones on a pallet	

4.6 Tests.

4.6.1 *Agent contamination (see 6.4).* The supplier shall submit to the Government his procedure for decontamination of filled and closed projectiles including his methods of controlling his process and shall obtain written approval from the Government prior to initiating the process as a permanent procedure. Subsequently, any permanent change he may make in his approved process must be approved in writing by the Government. Testing for the adequacy of the procedure shall be as specified in 4.5.3 and 4.6.1.1.

4.6.1.1 *Projectile.* The projectile shall be contaminated with liquid agent VX and the spot marked. The projectile shall be processed through the decontamination section of the filling line and then tested for agent contamination as follows:

The ticket (4.6.1.3) shall be removed from its envelope and the sensitive surface of the ticket wetted with two drops of buffer solution (4.6.1.5). The ticket shall be replaced in the envelope and the buffer solution kneaded into the ticket for approximately 15 seconds. The ticket shall be removed from the envelope and the sensitive surface of the ticket applied to the surface being checked for possible contamination. Sufficient finger pressure shall be applied to assure that firm contact of the entire sensitive surface area of the ticket is made with the surface of the projectile being checked. The ticket shall remain in contact with the surface being checked for at least 15 seconds. The ticket is again placed in the envelope and kneaded for 15 seconds. The ticket shall be removed from the envelope and two drops of substrate solution (4.6.1.4) shall be added to the ticket. The ticket shall be replaced in the envelope and kneaded. Within 90 seconds after the addition of substrate, the ticket should become blue if the area has been properly decontaminated.

4.6.1.2 *Closure.* The filled and closed projectile shall be placed on its side for a minimum of 4 hours, after which it shall be placed base down. A rubber stopper, machined to fit inside of the burster case, shall be inserted into the burster case forming a well (see 6.5). One hundred milliliters of buffer solution (4.6.1.5) shall be poured into the well formed by the rubber stopper in such manner that the closure is covered with solution. The solution shall be thoroughly stirred with a

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clean wooden applicator. The ticket (4.6.1.3) shall be immersed so the sensitive area is below the surface of the buffer solution and shall be held for 10 seconds without movement. The ticket shall be removed and placed in the envelope where it shall be kneaded for fifteen seconds. The ticket shall be removed from the envelope and two drops of substrate solution (4.6.1.4) shall be added to the ticket. The ticket shall be replaced in the envelope and again kneaded. If no agent is present on the projectile closure joint, the color of the ticket should turn blue within 60 seconds. If the color of the ticket is not blue, the ticket shall be placed in the envelope and kneaded for an additional 30 seconds. If the color is not blue this time, the projectile is considered contaminated and shall be rejected.

4.6.1.3 *Detector ticket.* The ticket shall be in accordance with MIL–D–51083.

4.6.1.4 *Substrate solution.* The substrate solution shall be in accordance with MIL–S–51189. It is not necessary to use the dispenser or to assemble the solution into the dispenser as specified in MIL–S–51189 for this application or may be prepared as follows:

The substrate solution shall be a 0.004 molar solution of 2,6 dichloroindophenyl acetate in ligroine (Eastman Kodak Co. No. P1628, D. = 0.72 – 0.74). Prepare the 2,6 dichloroindophenyl acetate by adding 32.6 grams of 2,6 dichloroindophenyl sodium salt in 30.6 grams of acetic anhydride. Shake on a mechanical wristaction shaker for 2 hours and allow to stand for 1 hour at room temperature. Pour onto crushed ice and after 1.5 hours. filter and wash the precipitate with water. Dissolve in diethyl ether, wash with water and dry over sodium sulfate. Concentrate the dried diethyl ether on a steam bath to about 10 ml, dilute with 4 volumes of ligroine (Eastman Kodak Co. No. P950 D. 0.64) until a cloudiness appears and filter. Cool the filtrate in a freezer and filter the crystalline ester. Recrystallize from diethyl ether—ligroine (P950). The recrystallized material shall have a melting point of 101° ± 1°C.

4.6.1.5 *Buffer solution.* The buffer solution shall be made up by dissolving 6.07 grams of tris (hydroxymethyl) amino methane in 250 milliliters (ml) of distilled water; 287 ml of 0.1 normal hydrochloric acid shall be added and diluted to 1 liter with distilled water. The pH shall be 8.0 ± 0.1.

4.6.2 *Helium leakage test.* Prior to closing, the void above the agent shall be evacuated to a pressure no greater than 7.5 pounds per square inch absolute (psia) and filled with helium to a maximum pressure of 22 psia to insure a partial pressure due to helium of no less than 15 pounds per square inch (psi) after closing. The closure area shall be sealed off by a test head manifolded to a vacuum system and a mass spectrometer type helium detector. The test head shall be calibrated to respond to a leak of 1 x 10^{-5} atmospheric cc per second of helium at the test head when a standard leak is introduced into the system at that point.

4.6.3 *Filling level.* The filling level shall be determined radiographically with the projectile oriented in a nose-up position. U. S. Army Edgewood Arsenal Inspection Standard R13–1 shall be used to determine conformance.

4.6.4 *Weight zone.* The projectile shall be weighed on a scale whose accuracy has been certified to one tenth of a pound.

5. PREPARATION FOR DELIVERY

5.1 **Packing, level A.** Unless otherwise specified in the contract or order, the filled and closed projectiles of one zone shall be packed for shipment by palletizing as specified on Drawing F7548346. The data card shall be in accordance with QAI 14–210–1.

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5.2 **Marking.** In addition to any special marking required by the contract or order, the palletized shipments shall be marked in accordance with Drawing F7548346 and MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The Government will furnish leak tester and radiographic equipment when required.

(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility when testing is performed by the Government.

(3) Packing and shipping instructions when applicable.

(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.2 **Filling aid.** When filled as specified in 3.3, the projectile carries approximately 14.82 pounds of VX. This information is furnished as an aid to procurement and not as a criterion for filling.

6.3 **Caution.** Due to the hazards of filling munitions with VX, suppliers are advised to exercise all necessary precautions to protect personnel from injury during filling and handling. Care should be taken to prevent wetting the closure with agent thereby making void the results of the helium test.

6.4 **Decontamination.** Decontamination can be accomplished by subjecting the munition to a spray of solution containing surface-active agent (e.g., "Nacconal" available from the Allied Chemical Dye Corporation, Baltimore, Maryland) for a minimum period of 1 minute in such a manner as to completely wash all surfaces. The munition should then be rinsed with water to remove all traces of the surface-active solution. The temperature of the spray and rinse water should be maintained at a range of 140° to 160°F. An approved procedure of control of the surface-active solution strength should be maintained by periodic examination and the time the munition is subjected to the spray should be automatically controlled and recorded. The contaminated overflow should be detoxified by chlorination of 2500 parts per million.

6.5 **Rubber stopper.** A periodic check should be maintained to check for contamination of the rubber stoppers. Stoppers found to be contaminated should be replaced.

Custodian:

Army — MU

Preparing activity:

Army — MU (EA)

Project No. 1320–A084

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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